

Exact Solution of the Pawn Game for up to Eight Pawns per Side

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Abstract

The pawn game—chess with pawns only—is solved exactly for $n = 1, \dots, 8$ pawns per side under two rulesets differing only in how a player with no legal move is scored: a *loss* (the stated rule) or a *draw* (chess stalemate). The game is finite; under the loss rule it is drawless, and the eight-pawn game is a first-player win, as are $n = 4, 6, 7$, while Black wins $n = 1, 2, 3, 5$. We give the outcomes, optimal openings, and winning technique—an unstoppable outside passed pawn—under both rulesets, and show en passant is essential: disabling it flips four of the sixteen values. The solver combines a bitboard search with a proved passed-pawn evaluator, board symmetries, a packed transposition table, and a lock-free parallel search, verified against an independent engine, unpruned search, and a Stockfish play-test the tablebase never loses. A pass variant—where a player may pass but not twice running—is a first-player win under the stated rule ($n \leq 8$) and a draw under chess stalemate ($n \leq 7$, conjectured beyond), showing outcomes governed by zugzwang.

Keywords

pawn game; exact game solving; combinatorial game theory; passed-pawn race; en passant; alpha-beta search; transposition table; chess endgames

1 Introduction

The pawn game was introduced to me by my chess coach, FIDE Master Carl Boor, who learned it from his father. It is a standard teaching position: two pawn phalanxes race toward promotion with no pieces to interfere. It isolates the pawn themes—passed pawns, breakthroughs, tempo, and zugzwang—that decide many endgames. Despite its simplicity, the game’s value beyond the smallest cases does not appear to have been settled in print: the four-pawn game is commonly cited as a White win (a standard breakthrough exercise), but we are aware of no published complete solution for larger n , and the eight-pawn game in particular has circulated as an open question in informal play (Pawn Game, n.d.; Lichess Forum, n.d.).

The game also has a motivation beyond itself. Philidor’s maxim that “the pawns are the soul of chess” (Philidor, 1749) takes the pawn structure to be a decisive factor even when pieces are present; to the extent one accepts that, the pawn game isolates a piece of full-chess reasoning—how a passed pawn is made and when a tempo wins the race—in a setting where it can be answered exactly. The transfer is only partial, since pieces change which breaks are available, but the motifs the solution exposes (the outside passer won by a tempo, the breakthrough sacrifice) are ones a player meets whenever pawns decide a game.

We determine the game-theoretic value of the n -versus- n pawn game for $n = 1, \dots, 8$ under perfect play, under both the game’s stated ruleset and the chess-stalemate variant, together with the optimal first moves and the winning lines. The contributions are: (i) the exact outcomes and openings for all $n \leq 8$, including the previously open eight-pawn game; (ii) a passed-pawn race evaluator whose every verdict is a

theorem, which under the loss rule settles a position outright and under the draw rule supplies two-sided bounds that drive alpha-beta cutoffs; (iii) the finding that en passant is essential—removing it changes four of the sixteen values, including every White win under the draw rule; and (iv) an open-source solver whose results are cross-checked against an independent move generator, against unpruned search, and against a strong external chess engine.

2 The game

The board is the 8×8 chessboard. White places n pawns on rank 2 and Black n pawns on rank 7, on the same n files (Figure 1). For $n < 8$ we take the files $a, \dots$ (left-justified); this is simply a labeling convention, so that the n -pawn game always uses the same n leftmost files. White moves first and the sides alternate. A pawn advances one square to an empty square, may advance two squares from its starting rank if both squares are empty, captures diagonally forward, and may capture en passant, exactly as in chess (Pawn Game, n.d.).

A player wins by (a) advancing a pawn to the opponent’s back rank (*touchdown*), (b) capturing the opponent’s last pawn, or (c) leaving the opponent with no legal move. Rule (c) is where the two rulesets diverge:

- **loss rule** (the stated rule): a player with no legal move loses;

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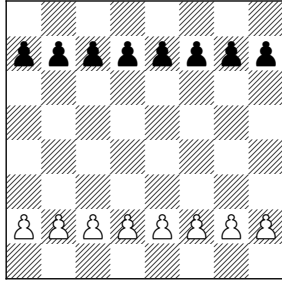


Figure 1. The eight-pawn starting position. White is to move. For $n < 8$ the two phalanxes occupy the leftmost n files.

- **draw rule** (chess stalemate): a player with no legal move draws.

Promotion choice is irrelevant: reaching the back rank ends the game at once. There are no kings, so no checks, pins, or castling arise.

3 Structure

Proposition 1. Finiteness. *Every game ends within 96 plies. The graph of reachable positions is a finite directed acyclic graph, and no position repeats.*

Proof. Every legal move advances exactly one pawn by one or two ranks toward promotion; captures advance the capturing pawn one rank. A pawn that starts on its own second rank reaches the far rank after at most six rank-advances, so each of the at most 16 original pawns makes at most six moves before it promotes (ending the game) or is captured. Hence at most 96 moves occur. The pawn count is non-increasing and, between captures, the sum of rank-advances strictly increases, so no position can recur; the position graph is acyclic.

Corollary 1. *Under the loss rule, the game has no draws: every position is a first-player win or a first-player loss.*

Proof. By Proposition 1 there are no infinite plays and no repetition draws, and the loss rule removes the only remaining non-decisive terminal. By Zermelo’s theorem (Zermelo, 1913), the finite two-outcome game of perfect information is determined.

Proposition 1 bounds the search depth and guarantees that plain memoized negamax terminates; Corollary 1 collapses the loss-rule search to a two-valued (win/loss) computation.

4 The solver

Representation. A position is two 64-bit pawn bitboards, the side to move, and the en-passant target (or none). Non-terminal positions keep pawns on ranks 2–7, so each bitboard uses 48 bits; the transposition key packs the two 48-bit masks, the side to move (1 bit), and a 5-bit en-passant code into 102 bits. The remaining bits of a 128-bit word carry a 2-bit value and a 2-bit bound flag, so a table entry is a single 128-bit integer with no separate key.

Search. We run fail-soft negamax alpha–beta. Values lie in $\{-1, 0, +1\}$; the root is searched with a wide window to return the exact value. The transposition table is open-addressing with linear probing; each entry records whether its value is exact or a lower/upper bound.

Terminals. A node returns without search when the side to move has no pawn (loss), the opponent has no pawn (win), a touchdown move exists (win), or no legal move exists (loss or draw by ruleset). These are the only rules the value depends on; everything else comes from the shortcuts of Section 5.

Implementation and hardware. The solver exists in two languages. A Python reference implementation (the oracle of Section 6) is written for clarity and validated move-for-move against `python-chess` (Fiekas et al., n.d.); the production solver is implemented in C++17. The two agree on all values and, at fixed settings, on exact node counts through $n = 6$. The transposition table is a flat open-addressing array of 2^k 128-bit words with linear probing; for the parallel search the words are 128-bit `std::atomic`. On the x86-64 target the compiler routes each 128-bit atomic compare-exchange through `libatomic` to the `cmpxchg16b` instruction, so a slot is updated in one indivisible step and is never read half-written, with no global lock. All runs reported here were on a single Google Cloud `n2-highmem-32` instance (32 vCPUs on Intel Cascade Lake, 256 GB of memory), compiled with `clang -O3 -mcx16 -latomic`; the parallel search used all 32 threads. The tables for $n \geq 5$ range from 2^{25} to 2^{33} entries (16-byte words, so 0.5 GB to 137 GB of allocated slots).

5 Pruning

5.1 A proved passed-pawn race evaluator

The evaluator decides a node from its pawn geometry alone—no subtree is expanded—and every verdict it returns is a theorem.

Definition 1. A pawn is an *unstoppable passer* if no enemy pawn stands on its file or on either adjacent file ahead of it, and no friendly pawn stands ahead of it on its own file (so its promotion path is clear). Such a pawn can be neither blocked nor captured, so by advancing it every move its owner promotes in a fixed number of that owner’s moves. Write a (resp. b) for the least such promotion distance over the side-to-move’s (resp. opponent’s) unstoppable passers, or ∞ if there are none. Write c_w (resp. c_b) for the least promotion distance over *all* of the side-to-move’s (resp. opponent’s) pawns ignoring obstacles—a lower bound on how soon that side could possibly promote.

Let plies be counted from the current position, so the side to move plays on odd plies (its k -th move at ply $2k - 1$) and the opponent on even plies.

Theorem 1. Loss rule. *Under the loss rule: if $a \leq c_b$ the side to move wins; if $b < c_w$ it loses.*

Proof. Suppose $a \leq c_b$. By advancing an unstoppable passer the side to move promotes at ply $2a - 1$. That passer cannot be captured, so the opponent can never remove all of its pawns and is never stalemated while it advances; the opponent’s only route to a win is its own promotion, needing at least c_b of

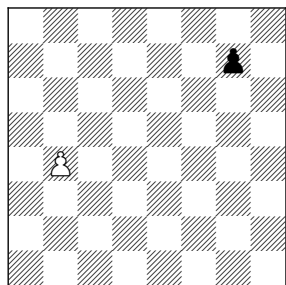


Figure 2. A position settled by Theorem 1 without search. White’s passer on b4 is unstoppable and promotes in four of White’s moves ($a = 4$); Black’s g-pawn needs at least five even using its double step ($c_b = 5$). Since $a \leq c_b$, White wins.

its moves, i.e. not before ply $2c_b \geq 2a > 2a - 1$. The side to move therefore wins first. The case $b < c_w$ is symmetric: the opponent promotes at ply $2b$; the side to move cannot capture the opponent’s passer or stalemate it, so its only route is promotion, needing at least c_w of its moves, i.e. not before ply $2c_w - 1 \geq 2b + 1 > 2b$.

The two conditions are mutually exclusive, since $a \leq c_b \leq b$ and $b < c_w \leq a$ are contradictory. The verdict is independent of the ruleset in the sense that neither side is stalemated in these lines, but the *loss-rule* reading of a stalemate is what makes the argument tight.

Corollary 2. Draw rule. *Under the draw rule: if $a \leq c_b$ then the value is ≥ 0 ; if $b < c_w$ then the value is ≤ 0 .*

Proof. The proof of Theorem 1 shows the opponent cannot win before the side to move promotes (case $a \leq c_b$), so the value is not a loss; under the draw rule a stalemate the opponent might reach is a draw, so the value is ≥ 0 rather than necessarily $+1$. The case $b < c_w$ is symmetric.

Under the loss rule, Theorem 1 yields exact leaves. Under the draw rule, Corollary 2 yields one-sided bounds around 0, injected into the search as window tightening: a lower bound of 0 raises α and cuts when $0 \geq \beta$; an upper bound of 0 lowers β and cuts when $0 \leq \alpha$. This is where the two rulesets differ in difficulty: under the loss rule a decided race removes a subtree; under the draw rule a stalemate is a resource that survives the bound, so the subtree is only narrowed.

Figure 2 shows a decided race. White’s b-pawn is an unstoppable passer four moves from promotion; Black has no passer and needs at least five moves to promote anything. By Theorem 1 White wins, and the node is settled without expanding it.

5.2 Symmetry

The game is invariant under mirroring the files and under swapping colors combined with a vertical flip; together these generate the game’s symmetry group, of order four (a vertical flip alone is not a symmetry, as pawns would move the wrong way). We fold each position to the least of its up-to-four images and key the transposition table by that representative.

5.3 Memory and parallelism

Two techniques trade computation for memory without affecting any value. First, terminal and race-decided nodes are recomputed rather than stored, since their cost is negligible; only searched nodes enter the table. Second, a node is stored only if its subtree exceeds a size threshold, so the many cheap subtrees are recomputed on demand; raising the threshold shrinks the table at the cost of more recomputation, which is how the eight-pawn tables are made to fit in memory.

The search is parallelized by splitting the tree below a fixed ply depth into independent subtree tasks distributed over a thread pool, all sharing one transposition table. Table slots are 128-bit atomics; on the target hardware these are lock-free, so no slot is ever read half-written. A race between threads can only recompute a subtree or replace one table entry with another that is itself valid, so scheduling cannot change the value the search returns; we confirm this empirically—the result is invariant to thread count through $n = 7$ (Section 6).

6 Verification

Correctness rests on four independent checks.

1. **Move generation.** The bitboard generator is fuzzed against `python-chess` on random kingless positions and random play-outs: the two agree on the legal-move set at every ply, covering en passant, double steps, captures, and touchdowns.
2. **Shortcut soundness.** The accelerated solver is compared against unpruned negamax on every position both evaluate, for both rulesets. This check found and removed an error: the race verdict is unsound under the draw rule (a stalemate defeats the race loss), which is why it is applied there only as a bound (Corollary 2).
3. **Implementation agreement.** The C++ core reproduces the reference solver’s values—and, at fixed settings, its exact node counts—through $n = 6$, and through $n = 7$ its value is invariant to every optimization flag (symmetry, color symmetry, subtree threshold, race evaluator, thread count), cross-checked against unpruned search where that is feasible. The independent Python oracle does not reach $n = 8$, so the eight-pawn values rest on this same solver, exhaustively validated through $n = 7$; we flag this as the point where independent corroboration is thinnest. As a further check, small-case draw-rule results with the pawns centered rather than left-justified reproduce values obtained independently from an earlier, separate implementation.
4. **Adversarial play-test (a one-sided check).** The tablebase plays Stockfish (Stockfish Developers, n.d.), used as a move evaluator (kings placed off the pawns’ path so it never moves a king), choosing a random optimal move each turn so the games differ. A perfect player cannot do worse than the game value against any opponent, so we play the tablebase only from the side it is not theoretically losing—the winner’s side when the game is decisive, White in a draw—and check that it never loses. This is a one-sided test: it would catch a value that is too optimistic (a claimed win or draw that is really lost must eventually lose a game) but not one that is too pessimistic; the exact correctness

argument rests on the first two checks. Table 1 reports 100 games against a depth-12 Stockfish for each n and rule (the $n = 8$ draw excepted, as its winning line coincides with the $n = 8$ loss line), all run on the same `n2-highmem-32` used for the solves. From a winning side the tablebase wins all 100 and from a drawing side it never loses; the loss column is zero in every row. The drawn rows carry a second lesson: from the drawing side the tablebase does not merely hold but wins most games—99 of 100 at $n = 7$ —because at a fixed search depth Stockfish cannot hold these theoretical draws. A general engine’s evaluation, built for positions with kings and pieces, is unreliable in kingless pawn races; only the exact table plays them correctly.

7 Results

Table 2 gives the value of the n -versus- n game under both rulesets, from White’s viewpoint. Under the loss rule, the sequence for $n = 1, \dots, 8$ is B, B, B, W, B, W, W, W : White wins for $n = 6, 7, 8$ and for $n = 4$. Under the draw rule, wins are rarer, and $n = 7$ differs between the rulesets—White’s seven-pawn win is by stalemate, which only the loss rule counts as a win.

Figure 3 shows the turning point of each of the eight games and how its value is decided under both rules.

Table 3 reports the size of each solve. All values, state counts, and wall times come from one setup—a Google Cloud `n2-highmem-32` instance (32 vCPUs, 256 GB), otherwise idle, each solve run once with all 32 threads.

The game is wide but shallow. A game lasts on the order of $4n$ plies: the mean depth rises from 3 plies at $n = 1$ to 34 ± 5 at $n = 8$, and the longest of two million random games was 57 plies—well inside the 96-ply worst-case bound of Proposition 1. The two axes scale very differently with n : the number of states grows exponentially (each added pawn multiplies the solved table by roughly an order of magnitude), while the depth grows only roughly linearly with n .

8 What it means to chess players

The solution has a consistent shape a player can use.

The winning method is an outside passed pawn the defender cannot answer. In every White win the plan is the same: a wing break manufactures a passed pawn on the flank that Black cannot stop, while White’s remaining pawns hold or capture Black’s counter-pawn—so White ends with an outside passer and Black with none. The decisive asymmetry is that only one side gets a passer; it is not a symmetric race in which both promote and White is a move quicker—in the lines below Black’s counter-runner is captured, not out-promoted. What tips the asymmetry to White is the tempo: moving first, and the tempo the wing break gains, let White be the side that emerges with the passer while its reserve pawns suffice to stop Black’s. The four-pawn win is the clearest miniature (Figure 4): 1.b4 a5 2.bxa5 b5 3.axb6 e.p. cxb6 4.a4, after which White’s b -pawn runs to $b7$ while $cxd3$ captures Black’s last counter-runner—so Black loses to the capture before the passer even needs to promote.

The winning opening is a flank break, not a central one. White’s winning first move is a wing double step that starts the outside passer: with four pawns it is $b4$ or $c4$, with six $c4$ or $d4$, with seven $c4$ or $e4$, and with eight $b4$ or $c4$ (with their mirror images $g4$, $f4$). On the full board the winning first moves happen to coincide with the flank openings— $c4$ (English) and $b4$ (Sokolsky)—though the resemblance is only the move itself: here White declines the center to promote a pawn on the b -file, not to fight for it with pieces. The eight-pawn win is the four-pawn miniature enlarged: 1.b4 a5 2.bxa5 b5 3.axb6 e.p. cxb6 4.a4 b5 5.axb5 d5 6.b6 d4 7.b7 d3 8.cxd3 e5 9.b8♯. There is no analogue here of the pawn structures that arise from an opening such as the Ruy Lopez; and where 1.e4 is Fischer’s famous “best by test” in chess, on the full board it does not win the pawn game at all—the only winning first moves are the flank breaks $b4$ and $c4$ (with mirrors $f4$, $g4$). The relevant reference point is the endgame breakthrough technique, executed here from the first move.

Material is subordinate to the passer. The perfect lines read strangely to a competitive player—pawns are given up freely—because the pawn game is decided by passers, not by a material count: what matters is which side ends with an unstoppable passed pawn, so a pawn spent to open a file or gain a tempo for the passer is spent well, exactly as in a king-and-pawn breakthrough. This is not the tablebase merely punishing weak defense. Against Stockfish, which defends soundly and does not shed material without cause, the winning side still wins every game (Table 1); and whenever White wins, Black is lost from move one—no Black reply does better—so the lines in Appendix A show optimal play by both sides in the game-theoretic (value-preserving) sense, not White exploiting a blunder. They are value-optimal, not tuned for longest resistance; the precise sense is given in Appendix A.

En passant is essential. The break in every White win in Appendix A is made with an en-passant capture: when Black answers a wing thrust with a two-square advance alongside, White takes en passant to open the file for the passer. To test whether this is a stylistic preference or a genuine requirement, we re-solved the whole game with en passant removed from the rules (Table 4). It is a requirement: removing en passant changes the value in four of the sixteen cases. The pattern is clean under the draw rule—there, every White win depends on en passant. White wins the draw-rule game at $n = 4, 6, 8$, and all three collapse to a draw once en passant is gone, because without it White cannot force the outside passer a tempo ahead and, with no stalemate to fall back on, has no other way through. Under the loss rule, White keeps its $n = 4, 6, 8$ wins without en passant: an alternative promotion break still gets through (the no-en-passant lines end in a touchdown), because a defender with no move now loses, so Black cannot fall back on the stalemate blockades that hold the draw-rule game. But the seven-pawn game still reverses outright, from a White win to a Black win. So en passant is not a stylistic flourish: at $n = 4, 6, 8$ (draw rule) and $n = 7$ (loss rule) it is the only route to White’s result, and taking it away changes who wins.

Comparing the two rules classifies each result. The rulesets differ only in whether having no move is a loss

n	rule	side	theory	W/D/L	table
1	loss	Black	+1	100/0/0	—
1	draw	White	0	0/100/0	—
2	loss	Black	+1	100/0/0	<1 KB
2	draw	Black	+1	100/0/0	<1 KB
3	loss	Black	+1	100/0/0	4.9 KB
3	draw	White	0	61/39/0	15 KB
4	loss	White	+1	100/0/0	160 KB
4	draw	White	+1	100/0/0	433 KB
5	loss	Black	+1	100/0/0	4.0 MB
5	draw	White	0	95/5/0	11.6 MB
6	loss	White	+1	100/0/0	89 MB
6	draw	White	+1	100/0/0	289 MB
7	loss	White	+1	100/0/0	2.0 GB
7	draw	White	0	99/1/0	6.9 GB
8	loss	White	+1	100/0/0	24.8 GB

Table 1. Tablebase versus Stockfish, 100 games for each n and rule, with the tablebase on the side it cannot theoretically lose from (the winner's side, or White in a drawn game). Stockfish is a move evaluator (never moving a king) searching to a fixed depth of 12 every move—a constant per-position budget; the tablebase plays a random optimal move each turn. “theory” is the game value from that side (in $\{-1, 0, +1\}$, with 0 a draw), “W/D/L” the results from the tablebase's viewpoint, and “table” the size of the perfect-play table it consulted (as in Table 3). The loss column is zero throughout: the tablebase wins every game from a winning side and never loses from a drawn one. The eight-pawn draw is not listed separately: White's winning line there is identical to the $n = 8$ loss promotion line (Appendix A).

n	loss rule	draw rule	White's winning openings
1	Black	$\frac{1}{2}$	—
2	Black	Black	—
3	Black	$\frac{1}{2}$	—
4	White	White	$b4, c4$
5	Black	$\frac{1}{2}$	—
6	White	White	$c4, d4$
7	White	$\frac{1}{2}$	$c4, e4$
8	White	White	$b4, c4$ (mirrors $g4, f4$)

Table 2. Game value of the left-justified n -versus- n pawn game (White to move) under both rulesets. The last column lists every first move that wins for White, where White wins; these are the same under both rulesets in the cases that White wins under both. For $n = 7$ White wins only under the loss rule, and the openings shown are its loss-rule winning moves. Throughout, $\frac{1}{2}$ denotes a draw.

n	loss rule			draw rule			depth (plies)	
	states	size	time (s)	states	size	time (s)	max	mean±std
1	0	—	< 0.01	0	—	< 0.01	4	3.0 ± 0.7
2	3	<1 KB	< 0.01	13	<1 KB	< 0.01	17	9.8 ± 3.3
3	309	4.9 KB	< 0.01	928	15 KB	< 0.01	26	14.2 ± 3.4
4	10,016	160 KB	< 0.01	27,085	433 KB	0.01	35	18.7 ± 3.5
5	251,538	4.0 MB	0.10	726,133	11.6 MB	0.36	40	22.9 ± 3.7
6	5,588,512	89 MB	2.32	18,037,094	289 MB	9.20	45	26.8 ± 4.1
7	123,981,255	2.0 GB	62.4	433,680,123	6.9 GB	234.5	51	30.5 ± 4.6
8	1,547,246,008	24.8 GB	873	3,611,840,263	57.8 GB	4681	57	34.1 ± 5.1

Table 3. Cached states, table size, and single-run wall time on the Google Cloud `n2-highmem-32` (32 vCPUs, 256 GB), every solve using 32 threads. Each stored position occupies exactly 16 bytes, so “size” is $16 \times$ the state count. State counts are the positions stored at the subtree-storage threshold in use (minsub = 16 throughout, except the draw-rule $n = 8$ at minsub = 64 so its table fits in memory); because the parallel solver inserts concurrently it stores a small fraction of positions more than once, so these counts slightly exceed the number of distinct positions and report the memory actually used rather than the full reachable set. Depth (plies from the start to the end of the game) is the same under both rules; “max” and “mean±std” are a Monte-Carlo estimate over 2×10^6 uniform-random games.

or a draw, so comparing them sorts the outcomes. Where the values agree— $n = 2$ (Black), and $n = 4, n = 6, n = 8$ (White)—the result is reached by promotion or capture and does not depend on the stalemate convention. Where they differ— $n = 1, 3, 5, 7$, each decisive under the loss rule but

drawn under chess stalemate—the loss-rule winner prevails only by forcing the opponent to have no legal move. The seven-pawn game is the sharpest case: White cannot force a promotion but can force Black into zugzwang, so White wins under the stated rule and only draws under chess

n	loss rule		draw rule	
	with e.p.	no e.p.	with e.p.	no e.p.
1	Black	Black	$\frac{1}{2}$	$\frac{1}{2}$
2	Black	Black	Black	Black
3	Black	Black	$\frac{1}{2}$	$\frac{1}{2}$
4	White	White	White	$\frac{1}{2}$
5	Black	Black	$\frac{1}{2}$	$\frac{1}{2}$
6	White	White	White	$\frac{1}{2}$
7	White	Black	$\frac{1}{2}$	$\frac{1}{2}$
8	White	White	White	$\frac{1}{2}$

Table 4. Game value with and without en passant, under both rules. Removing en passant flips four values (bold): every White win under the draw rule ($n = 4, 6, 8$, each White→draw) and the seven-pawn game under the loss rule (White→Black). Everywhere else White either has an alternative winning break (its $n = 4, 6, 8$ loss-rule wins still promote without en passant) or was not winning, so the value is unchanged.

stalemate. The choice of rule matters, and it matters exactly at $n = 1, 3, 5, 7$.

Symmetry breaks in White's favor once the board is full enough. For small n , Black holds by mirroring; the first-move advantage is not enough to break a short, thin phalanx. From $n = 6$ on (and already at $n = 4$), White has the space to force a break Black cannot answer, and the mirror fails.

9 A third rule: passing

A move in a real game rarely forces a change in the pawn structure. With pieces on the board a player can almost always make a waiting move, and a stalemate is a draw. How much of the pawn game's outcome, then, is really about *zugzwang*—being forced to move—rather than about the pawns themselves? To find out we solved a third variant in which a player may, on their turn, either move a pawn or *pass*. Two passes in a row would let the game run forever, so a pass is legal only when the opponent did not just pass. That single restriction keeps the game a finite acyclic graph—a pass cannot chain, and every real (pawn) move still makes the irreversible progress of Proposition 1 (the pawn count falls or the total rank-advance rises)—so the same solver applies, with one extra bit of state (may the side to move pass?).

The answer is clean and constant in n across the solved range:

- under chess stalemate, *every* game is a **draw** (solved exhaustively for $1 \leq n \leq 7$);
- under the game's stated (no-move-loses) rule, *every* game is a **first-player (White) win** (solved exhaustively for $1 \leq n \leq 8$).

The one value we did not compute is the eight-pawn draw. Under the draw rule the search stores far more of the game tree than under the loss rule—at $n = 7$, 1,231,693,353 states against 386,934,545—so an eight-pawn draw-rule table runs to well over ten billion positions. Our transposition table is open-addressed and sized to a power of two above the live-entry count (to keep the load factor below one); storing that many entries needs a 2^{34} -slot table, which at 16 bytes per slot is 275 GB and exceeds the 256 GB machine we used, even at an aggressive storage threshold. Because the value is constant in n across the entire range we did solve, we record the eight-pawn draw as a conjecture rather than a theorem. So

the pawn game's decisions are, at bottom, a matter of tempo. The breakthroughs of Section 8—including the en-passant wins at $n = 4, 6, 8$ —exist only because the defender is *forced* to advance a pawn into the capture; give the defender a free waiting move and the breakthrough evaporates (chess rule), while conversely the mover's power to hand the opponent the move becomes decisive (the stated rule turns every position into a first-player win).

This connects to over-the-board play, with a caveat. When kings are present a player usually *does* have a spare tempo—a king move that leaves the pawns untouched—and a stalemate really is a draw, so of our three variants the chess-stalemate-with-passing model is the closest to a king-and-pawn ending. The match is only partial: a pass is an unlimited waiting resource, whereas a king holds only finitely many spare tempi and must eventually commit; and a real king does far more than waste a move—it blocks the outside passer, captures and defends pawns, and escorts its own. The model isolates the tempo role and discards the king's activity. Within that idealization its verdict is that no *balanced, symmetric* structure of the kind studied here—up to eight pawns per side—yields a forced breakthrough against a defender who can wait: the winning breaks in the forced variants are *zugzwang* breaks, not structural ones. (Asymmetric structures, such as the classic majority whose breakthrough wins regardless of the move, are structural and lie outside this model.) A healthy, non-committal structure holds as long as the defender keeps a tempo in hand; the practical lesson is the familiar one that in pawn play the side that runs out of useful waiting moves first is the one that must create weaknesses.

The pass variant was solved on the same machine and checked the same way. Its values match an independent memoized-negamax reference (which uses no symmetry or pruning) through $n = 6$, so the passed-pawn race evaluator remains sound with passing added; and the tablebase never loses to Stockfish across 100 games per n and rule (100 games at each size and rule, depth-12 Stockfish; the tablebase never lost—loss rule through $n = 7$, draw rule through $n = 6$; Stockfish has no pass move, which only makes it a weaker opponent). The eight-pawn loss-rule pass table is of the same order as the base game: 3,752,595,781 states (60 GB, solved in 110 minutes).

Patterns, and beyond eight pawns. A reader will look for a pattern in the value sequences. Under chess stalemate

the base game has a clean one: every odd n is a draw and every even n is a White win except $n = 2$ (Black). Under the stated rule White wins for $n = 4$ and every $n \geq 6$. We can offer no proof that either pattern continues, and—as is typical of combinatorial games—there may be no closed form at all; with no data beyond $n = 8$, any extrapolation to $n \geq 9$ is speculation. The pass variant is the exception: its values are constant in n over the solved range (all draws under chess stalemate, all first-player wins under the stated rule), which at least makes a uniform continuation the natural conjecture, though still only a conjecture.

10 Conclusion

The pawn game is solved for up to eight pawns per side under both rulesets. The eight-pawn game is a first-player win under the game’s stated rule and under chess stalemate alike, and every White win is the same outside-passer race. The solver’s engine—a proved race evaluator, the game’s symmetry group, a packed atomic transposition table, and a lock-free parallel search—settles the largest case on one multicore machine. A third rule—allowing a waiting move (a pass)—shows the outcomes to be governed by zugzwang in this kingless setting: with passing the game is a draw under chess stalemate (solved for $n \leq 7$, conjectured beyond) and a first-player win under the stated rule (solved for $n \leq 8$). Open directions include other starting placements, larger boards, and whether these patterns persist beyond eight pawns per side.

Availability

The solver (Python reference oracle and C++ core), the test suite, an engine API for playing the solved tablebase as a perfect kingless player, and the raw result files behind every table in this paper are openly available, with build and reproduction instructions, at <https://github.com/orenlivne/pawngame> (Livne, 2026).

Acknowledgments

I am grateful to FIDE Master Carl Boor for introducing me to the pawn game and for valuable correspondence, including the questions that prompted the pass variant of Section 9 and the discussion of patterns.

A Principal variations

Each line is a perfect game from the left-justified start under optimal play by both sides. Here “optimal” means value-preserving: every move keeps the position’s game-theoretic value, so the winner retains the win and the loser can do no better (from a lost position every legal move loses equally). These are therefore value-optimal lines rather than lines chosen to maximize the loser’s resistance; among value-equal moves the solver follows a fixed ordering, so each line is one representative of perfect play, not the unique or the longest one. Notation: a straight push is written as its destination square; a diagonal move is a capture $\langle \text{file} \rangle x \langle \text{dest} \rangle$, with *e.p.* for en passant; ‡ marks the touchdown that ends the game. White’s moves are numbered. A line that ends without a ‡ terminates either by the capture of the opponent’s last pawn (a win under both rulesets) or by the side to move having no

legal move (a stalemate, scored by the ruleset in force); the value in Table 2 gives the outcome.

Loss rule

- $n = 1$ 1. a4 a5
- $n = 2$ 1. a4 a5 2. b4 axb4 3. a5 b3 4. a6 bxa6
- $n = 3$ 1. a4 a5 2. b4 axb4 3. a5 b3 4. cxb3 c5 5. a6 bxa6 6. b4 cxb4
- $n = 4$ 1. b4 a5 2. bxa5 b5 3. axb6 e.p. cxb6 4. a4 b5 5. axb5 d5 6. b6 d4 7. b7 d3 8. cxd3
- $n = 5$ 1. a4 c5 2. a5 c4 3. a6 bxa6 4. b4 cxb3 e.p. 5. cxb3 a5 6. d4 a4 7. bxa4 d5 8. a5 a6 9. e4 dxe4 10. d5 e3 11. d6 exd6
- $n = 6$ 1. c4 a5 2. c5 a4 3. d4 a3 4. bxa3 b5 5. cxb6 e.p. cxb6 6. d5 b5 7. d6 exd6 8. e4 b4 9. axb4 d5 10. exd5 f5 11. d6 f4 12. b5 f3 13. b6
- $n = 7$ 1. c4 a5 2. d4 a4 3. b4 axb3 e.p. 4. axb3 b5 5. cxb5 c5 6. bxc6 e.p. dxc6 7. d5 cxd5 8. f4 d4 9. f5 d3 10. exd3 e5 11. fxe6 e.p. fxe6 12. g4 e5 13. g5 e4 14. dxe4 g6 15. e5
- $n = 8$ (via 1. c4, the primary line) 1. c4 a5 2. d4 a4 3. b4 axb3 e.p. 4. axb3 b5 5. cxb5 c5 6. bxc6 e.p. dxc6 7. d5 cxd5 8. f4 d4 9. f5 d3 10. exd3 e5 11. fxe6 e.p. fxe6 12. g4 e5 13. g5 e4 14. dxe4 h5 15. gxh6 e.p. gxh6 16. e5 h5 17. e6 h4 18. e7 h3 19. e8‡
(via 1. b4) 1. b4 a5 2. bxa5 b5 3. axb6 e.p. cxb6 4. a4 b5 5. axb5 d5 6. b6 d4 7. b7 d3 8. cxd3 e5 9. b8‡

Draw rule

- $n = 1$ 1. a4 a5
- $n = 2$ 1. a4 a5 2. b4 axb4 3. a5 b3 4. a6 bxa6
- $n = 3$ 1. a4 a5 2. c3 b6 3. b4 c5 4. bxa5 bxa5 5. c4
- $n = 4$ 1. b4 a5 2. bxa5 b5 3. axb6 e.p. cxb6 4. a4 b5 5. axb5 d5 6. b6 d4 7. b7 d3 8. cxd3
- $n = 5$ 1. b4 d5 2. b5 d4 3. a4 a5 4. bxa6 e.p. bxa6 5. a5 d3 6. cxd3 c5 7. e4 c4 8. dxc4 e5 9. c5
- $n = 6$ 1. c4 a5 2. d4 a4 3. b4 axb3 e.p. 4. axb3 b5 5. cxb5 c5 6. bxc6 e.p. dxc6 7. d5 cxd5 8. f4 d4 9. f5 d3 10. exd3 e5 11. fxe6 e.p. fxe6 12. b4 e5 13. b5 e4 14. dxe4
- $n = 7$ 1. a4 c5 2. e4 e5 3. a5 c4 4. g3 b6 5. axb6 axb6 6. f4 f5 7. exf5 exf4 8. gxf4 b5 9. f6 gxf6 10. f5 c3 11. bxc3 d5 12. d4 b4 13. cxb4
- $n = 8$ 1. b4 a5 2. bxa5 b5 3. axb6 e.p. cxb6 4. a4 b5 5. axb5 d5 6. b6 d4 7. b7 d3 8. cxd3 e5 9. b8‡ (the 1. b4 promotion line; identical to the loss-rule line, since it wins by touchdown rather than by stalemate)

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The maxim “the pawns are the soul of chess” appears on p. xix (original “*les Pions. . . ils sont l’âme des Echecs*”; rendered in the 1750 English edition as “the Pawns: they are the very life of this game”).
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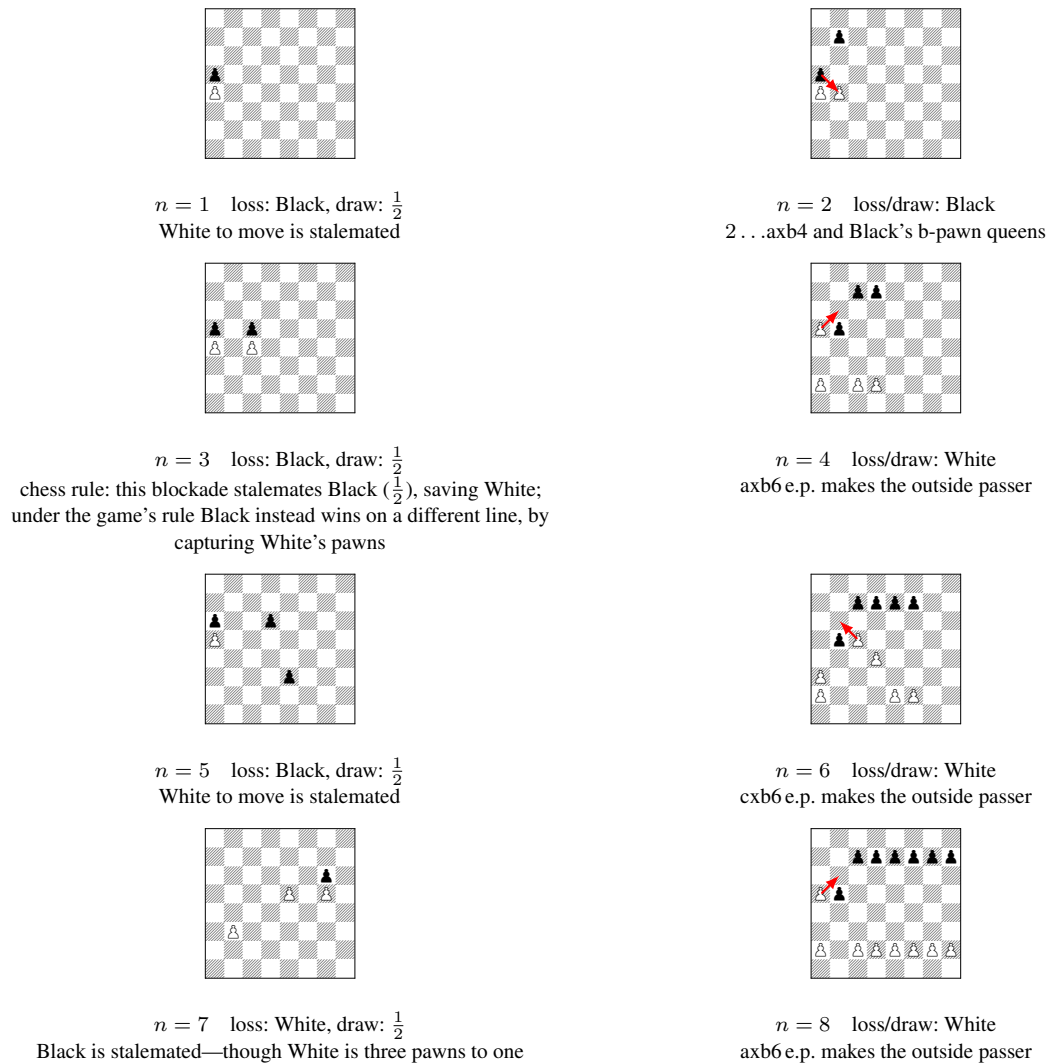


Figure 3. The turning point of each game, taken from the perfect line. **Right column (even n):** the decisive capture (red arrow). For $n = 4, 6, 8$ White wins with an en-passant break (axb6 or cxb6) that manufactures an outside passed pawn; for $n = 2$ White's only break rebounds and Black's own b-pawn queens. **Left column (odd n):** the result turns on a stalemate—the side to move has no legal move, which the game's rule scores as a loss and the chess rule as a draw. This is why the four odd values differ between the rulesets while the four even values, won by promotion or capture, agree. In the $n = 7$ frame White is a pawn phalanx up yet under the chess rule wins nothing: it can only stalemate Black. For $n = 3$ the stalemate is instead White's drawing resource under the chess rule, while under the game's rule Black wins by leaving White with no pawns.

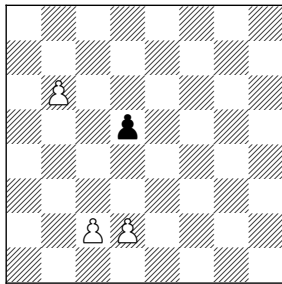


Figure 4. The asymmetry that wins. Position from the four-pawn game after 1.b4 a5 2.bxa5 b5 3.axb6 e.p. cxb6 4.a4 b5 5.axb5 d5 6.b6. White's *b*-pawn is passed—no Black pawn stands on the *a*-, *b*-, or *c*-file to stop it—and is unstoppable (*b7*). Black's *d*-pawn is *not* passed: White's reserve *c2* and *d2* hold it (6...d4 7.b7 d3 8.cxd3), so Black loses its last pawn before the *b*-pawn even needs to promote and never gets a passer of its own. This is the winning plan throughout: sacrifice a wing pawn to make a single outside passer while the remaining pawns neutralize Black's counter-pawn—White ends with one passer, Black with none.